

ECEN-601: Homework 5

Due on October 28, 2025 at 11:59 pm

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Problem 1

Let V be a vector space over F . Show that the sum of two inner products on V is an inner product on V . Is the difference of two inner products an inner product? Show that a positive multiple of an inner product is an inner product.

Solution

Problem 2

Let V be a real or complex vector space with an inner product. Show that the quadratic form determined by the inner product satisfies the *parallelogram law*

$$\|\alpha + \beta\|^2 + \|\alpha - \beta\|^2 = 2\|\alpha\|^2 + 2\|\beta\|^2.$$

Solution

Problem 3

Show that the formula

$$\left\langle \sum_j a_j x^j \mid \sum_k b_k x^k \right\rangle = \sum_{j,k} \frac{a_j b_k}{j+k+1}$$

defines an inner product on the space $R[x]$ of polynomials with real coefficients. Let W be the subspace of polynomials of degree less than or equal to n . Find the weight matrix associated with this inner product on W , relative to the ordered basis $1, x, x^2, \dots, x^n$. (*Hint:* To show that the formula defines an inner product, observe that

$$\langle f \mid g \rangle = \int_0^1 f(t)g(t) dt$$

and work with the integral.)

Solution

Problem 4

Let V be the vector space of real polynomials with degree at most 3. Equip V with the inner product

$$\langle f | g \rangle = \int_0^1 f(t)g(t) dt.$$

- (a) Find the orthogonal complement of the subspace of constant polynomials.
- (b) Apply the Gram-Schmidt process to the basis $\{1, x, x^2, x^3\}$.

Solution

Problem 5

Let V be the vector space of all $n \times n$ matrices over C , with the inner product $(A|B) = \text{tr}(AB^H)$. Find the orthogonal complement of the subspace of diagonal matrices.

Solution

Problem 6

Let W be a finite-dimensional subspace of an inner product space V , and let E be the orthogonal projection of V in W . Prove that $\langle E\alpha|\beta\rangle = \langle\alpha|E\beta\rangle$ for all α, β in V .

Solution